

CSE4261: Neural Network and Deep Learning

Assignment-1

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Accuracy Comparison Table

Table 1: Transfer Learning vs Fine-tuning Accuracy (Softmax Activation)

Model	Transfer Accuracy	Fine-tune Accuracy
MobileNet	0.7355	0.7415
VGG16	0.6040	0.7955
ResNet50	0.0850	0.1290
InceptionV3	0.6075	0.6540
InceptionResNetV2	0.6685	0.6755
MobileNetV2	0.7195	0.7360
DenseNet121	0.7270	0.7290
XceptionNet	0.6875	0.6990
NasNetMobile	0.6425	0.6415
DenseNet201	0.7440	0.6085

Comparative Analysis

We evaluated the performance of ten popular convolutional neural network (CNN) architectures on a subset of the CIFAR-100 dataset, specifically focusing on 20 fine-label classes. The evaluation was done in two phases: transfer learning and fine-tuning, using Softmax as the final activation function.

From the results, we observe that **VGG16** achieves the highest fine-tuning accuracy (0.7955), despite a relatively low transfer learning accuracy (0.6040). This indicates that VGG16 benefits significantly from fine-tuning, suggesting that its deeper layers adapt well to new data distributions.

MobileNet and **MobileNetV2** also perform well, with both achieving fine-tuning accuracies above 0.73. These lightweight models not only offer high accuracy but are also computationally efficient, making them excellent choices for embedded or mobile applications.

DenseNet201 exhibits the highest transfer accuracy (0.7440) but surprisingly drops during fine-tuning to 0.6085. This suggests that while its pre-trained features generalize well, further training on limited data may lead to overfitting or instability in deeper layers.

ResNet50 is an outlier, performing poorly in both transfer (0.0850) and fine-tuning (0.1290). This could be due to vanishing gradient issues or the dataset-specific characteristics that ResNet50 fails to capture effectively without more aggressive augmentation or hyperparameter tuning.

XceptionNet and **InceptionResNetV2** demonstrate balanced performance in both transfer and fine-tuning phases, confirming their robust architecture built on depthwise separable convolutions and residual learning, respectively.

NasNetMobile shows moderate performance but no significant improvement in fine-tuning, implying that its search-optimized architecture might be more sensitive to the dataset size or label granularity.

Overall, models like **VGG16**, **MobileNet**, and **DenseNet121** stand out for their strong fine-tuning results. In contrast, ResNet50 underperforms in this experimental setting, suggesting that architecture suitability varies significantly with dataset scale and complexity.

Code Link:

Here's the github link of the implemented code: <https://github.com/mdsajjadhossain25/CSE4261>